AI Virtual Mouse using Hand Gestures

A computer vision system that transforms hand movements into cursor control, replacing traditional mouse interaction with intuitive gestures captured via webcam.

Project Overview

Core Concept

This innovative system leverages computer vision and AI-based hand tracking to create a seamless virtual mouse experience. Users control cursor movement and perform clicks entirely through natural hand gestures, opening possibilities for touchless interaction in various environments.

Primary Applications

- Accessibility solutions for users with mobility challenges
- Touchless interfaces for sterile environments
- Interactive presentations and

Key Features

Real-time Tracking

Continuous hand detection with minimal latency using standard webcam hardware

Cursor Control

Precise movement controlled by index finger positioning and tracking

Click Action

Intuitive pinch gesture between thumb and index finger triggers mouse clicks

Responsive Design

Smooth, natural interaction with optimized frame processing rates

Technology Stack

1

Python 3.10

Core programming language with optimal MediaPipe compatibility for stable landmark detection

2

OpenCV

Real-time image processing framework handling video capture and frame manipulation

3

MediaPipe

Machine learning pipeline for precise 21-point hand landmark detection and tracking

4

PyAutoGUI

System-level mouse control library translating gestures into cursor movements and clicks

5

NumPy

Mathematical computations for distance calculations between finger landmarks

How It Works

01

Video Capture

Webcam streams live video frames continuously processed at approximately 30 FPS

02

Landmark Detection

MediaPipe identifies 21 precise anatomical points on the hand including fingertips, knuckles, and wrist

03

Cursor Mapping

Index fingertip coordinates are translated to screen positions using proportional scaling

04

Distance Calculation

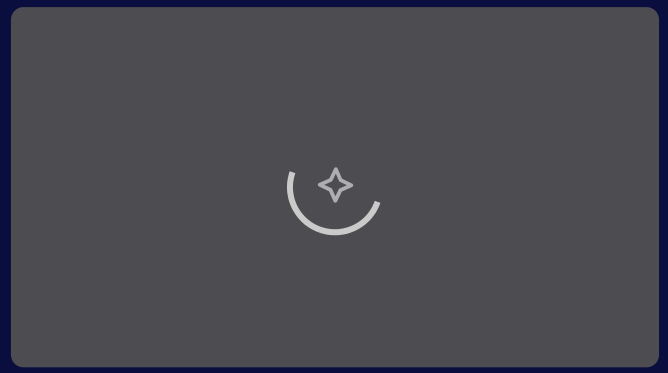
Euclidean distance between thumb and index finger measured in real-time using NumPy

05

Action Trigger

Small distance threshold (<50 pixels) activates left click; larger distances enable movement only

Gesture Controls Reference



Move Cursor

Position index finger and move hand across camera view to control cursor position on screen

Left Click

Bring thumb and index finger together in pinching motion to execute mouse click action

No Action

Remove hand from camera view to pause cursor control and prevent unintended movements

System Requirements

Hardware Requirements

- Built-in or external webcam (minimum 640×480 resolution)
- Processor: Modern CPU (Intel i3 or equivalent minimum)
- RAM: 4GB minimum, 8GB recommended for optimal performance

Software Requirements

- Python 3.10 (recommended for MediaPipe compatibility)
- OpenCV library (cv2) for image processing
- MediaPipe framework for hand tracking
- PyAutoGUI for mouse control functions
- NumPy for mathematical operations

Environmental Requirements

- Adequate lighting conditions (avoid backlighting)
- Clear hand visibility without obstructions
- Stable camera position with minimal movement
- Uncluttered background for better detection

Limitations & Considerations

Lighting Dependency

Requires consistent, stable lighting conditions. Performance degrades significantly in low light or harsh shadows

Camera Quality Impact

Lower resolution cameras reduce tracking precision and responsiveness. Higher frame rates improve user experience

Python Version

MediaPipe currently optimized for Python 3.10. May encounter compatibility issues with newer

Usage Context

Best suited for controlled environments with consistent lighting and minimal background interference. Not recommended for highly variable lighting conditions

Future Development Roadmap

Extended Gesture Set

Implement right-click functionality, scroll wheel emulation, and drag-and-drop operations through additional gesture recognition

System Controls

Add volume adjustment, media playback controls, and brightness management using intuitive hand gestures

Multi-hand Support

Enable simultaneous tracking of both hands for complex interactions and advanced gesture combinations

GUI Interface

Develop intuitive graphical interface for sensitivity adjustment, gesture calibration, and system configuration

Project Details

Implementation

Developed in Python 3.10 using OpenCV for real-time video processing, MediaPipe for hand landmark detection, and PyAutoGUI for system-level mouse control.

Running the Project

```
python main.py
```

Execute the main script to start the virtual mouse system. The webcam will activate automatically and begin tracking hand gestures.

Project Author

Saumya Rani